
ECE228 Project Proposal: Solar Power Generation Prediction

Zichen Hu(zih050@ucsd.edu)

Xiaobo Li(xil341@ucsd.edu)

Zheng Dong(z6dong@ucsd.edu)

(Team 15)

1 Project Proposal: Track 2

- **Problem Background & Motivation:**

Solar power is one of the most widely used renewable energy sources, and its role in the energy system is becoming more important as more countries try to reduce the use of fossil fuels. However, solar power generation is not as controllable as traditional power. The output of a solar plant can change with weather conditions, cloud cover, solar radiation, temperature, and the position of the sun. Due to this variability, it can be difficult for the power grid to estimate how much solar energy will be available at a certain time.

Accurate solar power prediction is important because electricity supply and demand need to be balanced. If solar generation can be predicted more reliably, people can make better decisions about energy storage, backup generation and power distribution. For example, when the model predicts low solar output, the system can prepare other energy sources earlier. When high solar output is expected, we can make better use of the energy instead of wasting it. With better prediction, we can improve energy scheduling.

In this project, we will study the topic using meteorological and solar-position data. The goal is to predict the generated solar power based on a variety of features. In addition to building prediction models, we will also examine which features have the strongest effect on solar power output and compare several machine learning methods to see which model(s) are most suitable for this problem.

- **Related Work:**

Previous work on our topic has shown that photovoltaic(PV) output is strongly affected by environmental and meteorological conditions. Variables such as solar radiation, temperature, humidity, wind speed, and cloud cover can cause solar power output to fluctuate. Tripathi et al. studied PV panel power prediction under changing environmental conditions and compared multivariate regression, support vector machine regression, and Gaussian regression. Their results showed that SVM regression performed better than the other models, suggesting that nonlinear regression methods can be effective for solar power prediction.

More recent research has explored advanced machine learning and deep learning methods for solar energy forecasting. Shakhovska et al. used detailed meteorological data and compared models such as SVM, KNN, CatBoost, LSTM and LSTM with inception layers. Their study also emphasized preprocessing, feature selection and restricting predictions to daylight hours. Similarly, Rizk-Allah et al. proposed an LSTM-based forecasting model to

identify important environmental factors. These works show that model comparison and feature interpretation are both important parts of solar power prediction.

Here are some of the papers that we use for this project:

Tripathi et al.

Shakhovska et al.

Rizk-Allah et al.

- **High-level Methodology:**

For the methodology, we plan to treat solar power generation prediction as a supervised regression problem. The target variable will be the generated solar power, and the input variables will include meteorological and solar-position features. We will first clean the dataset and check for missing values and outliers. We will also perform exploratory data analysis to understand the distribution of the target variable and the relationship between each feature and generated power.

After preprocessing the data, we will split the dataset into training and testing sets and build several prediction models. We plan to start with simple baseline models like linear regression, then compare them with nonlinear machine learning models such as random forest regression, neural network, gradient boosting etc. The models will be evaluated using regression metrics such as mean absolute error, root mean squared error and R^2 . This comparison will help determine which type of model is more suitable for predicting solar power generation from environmental features.

Besides prediction accuracy, we will also analyze the most influential factors in the model. For tree-based models, we can use feature importance scores to identify which variables contribute most to the prediction. We may also compare these results with correlation analysis to see whether the important features match physical expectations. The final goal is to build a model that can reasonably predict solar generation while also giving some explanation of which weather and solar-position factors are most important.